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A Survey of Context Engineering for Large Language Models

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The performance of Large Language Models (LLMs) is fundamentally determined by the contextual information provided during inference. This survey introduces Context Engineering, a formal discipline that transcends simple prompt design to encompass the systematic optimization of information payloads for LLMs. We present a comprehensive taxonomy decomposing Context Engineering into its foundational components and the sophisticated implementations that integrate them into intelligent systems. We first examine the foundational components: context retrieval and generation, context processing and context management. We then explore how these components are architecturally integrated to create sophisticated system implementations: retrieval-augmented generation (RAG), memory systems and tool-integrated reasoning, and multi-agent systems. Through this systematic analysis of over 1400 research papers, our survey not only establishes a technical roadmap for the field but also reveals a critical research gap: a fundamental asymmetry exists between

model capabilities. While current models, augmented by advanced context engineering, demonstrate remarkable proficiency in understanding complex contexts, they exhibit pronounced limitations in generating equally sophisticated, long-form outputs. Addressing this gap is a defining priority for future research. Ultimately, this survey provides a unified framework for both researchers and engineers advancing context-aware AI.

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